

$$x_3 = (1) \pmod{6}$$

$$x_0 = (1, 3, 5, 7)$$

$$ax \equiv b \pmod{m}$$

$$x_0 \equiv x_1 \pmod{m}$$

$$ax \equiv b \pmod{b}$$

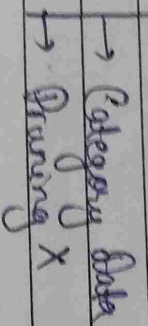
$$ax_2 \equiv ax_1 \pmod{m}$$

$$ax_2 \equiv b \pmod{m}$$

Decision Trees

3 Methods

→ Information Gain (ID3) → (Iterative Dichotomiser 3)



→ Gain Ratio (4-5 or Improvement of ID3)

$$E = - \sum_{i=1}^n p_i \log p_i$$

IG = E(d) - Weighted of Entropy node

Treasury Outlook

→ Calculate $P(\text{Yes})$ & $P(\text{No})$

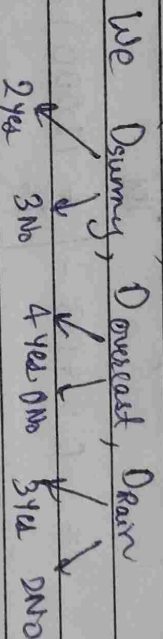
→ Step 1 (Base Entropy $H(D)$)

$$H(D) = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} \\ = 0.9403$$

→ Step 2: Information gain for all attributes

Now we have to choose a root node for that we'll calculate IG for each attribute

→ Gain (Outlook)



$$H(D_{\text{Sunny}}) = -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} = \frac{0.5288 + 0.4422}{2} = 0.9710$$

$$H(D_{\text{Overcast}}) = -\frac{4}{4} \log_2 \frac{4}{4} - 0 = 0.0000$$

$$H(D_{\text{Rain}}) = -\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5} = \frac{0.9403 + 0.6931}{2} = 0.9710$$

$$\text{Gain}(\text{Outlook}) = \frac{H(D)}{101} = \sum \frac{10 \cdot 1}{101} H(D_i) \\ = 0.9403 - \left(\frac{5}{14} \times 0.9710 + \frac{4}{14} (0.0000) + \frac{5}{14} (0.9710) \right)$$

$$= 0.2467$$

(c) Gain (Temperature)

$$\begin{aligned} D_{\text{hot}} &\rightarrow 2Y, 2N \\ D_{\text{mid}} &\rightarrow 4Y, 2N \\ D_{\text{cool}} &\rightarrow 3Y, 1N \end{aligned}$$

$$H(D_{\text{hot}}) = -\frac{2}{4} \log_2 \frac{2}{4} - \frac{2}{4} \log_2 \frac{2}{4} = 1$$

$$H(D_{\text{mid}}) = -\frac{4}{6} \log_2 \frac{4}{6} - \frac{2}{6} \log_2 \frac{2}{6} = 0.918$$

$$H(D_{\text{cool}}) = 0.8113$$

$$\text{Gain (Temperature)} = H(0) - \sum \frac{D_i}{|D|} H(D_i)$$

$$= 0.9403 - \left[\frac{4}{14} (1.000) + \frac{6}{14} (0.918) + \frac{4}{14} (0.8113) \right]$$

$$= 0.0293$$

Gain (Humidity)

$$D_{\text{high}} \rightarrow 3Y, 4N, \quad D_{\text{normal}} = 6Y, 1N$$

$$H(D_{\text{high}}) = -\frac{3}{7} \log_2 \frac{3}{7} - \frac{4}{7} \log_2 \frac{4}{7}$$

$$= 0.5238 + 0.4628$$

$$= 0.9852$$

$$H(\text{Outlook}) = -\frac{6}{7} \log_2 \frac{6}{7} - \frac{1}{7} \log_2 \frac{1}{7} =$$

$$= 0.1918 + 0.4011$$

$$= 0.5917$$

$$\text{Gain (Humidity)} = 0.9403 - \left[\frac{7}{14} (0.9852) + \frac{7}{14} (0.5917) \right]$$

$$= 0.9403 - 0.7885$$

$$= 0.1518$$

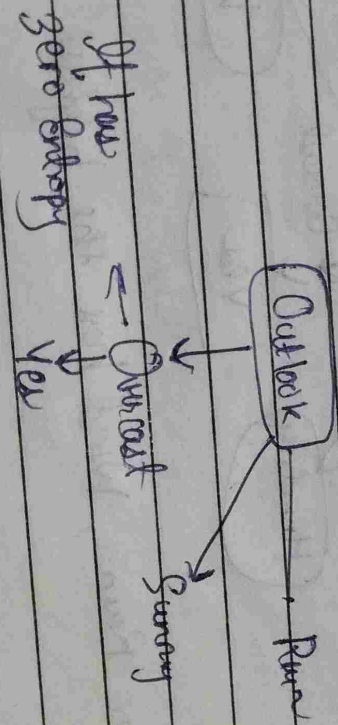
Similarly Grain (Wind)

$$\text{Gain (Wind)} = 0.9403 - \left[\frac{8}{14} (0.8113) + \frac{6}{14} (0.0000) \right]$$

$$= 0.9403 - 0.8922$$

$$= 0.0481$$

Since Outlook has the highest information gain
Outlook is the best node.



For sunny (Outlook) & Humidity

$\begin{matrix} & \swarrow & \searrow \\ 3H & & 2N \\ 01 & & 14 \end{matrix}$

$$IG = 0.9710 - \left[\frac{3(0) + 2(0)}{5} \right] = 0.9710$$

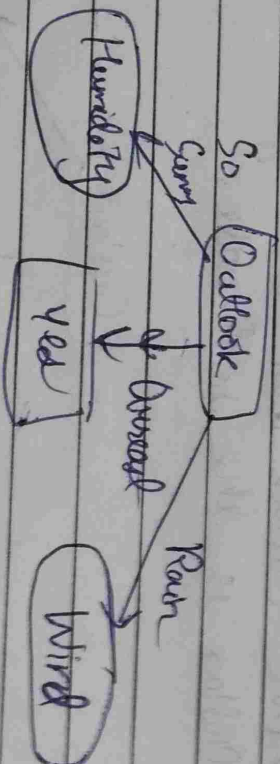
Now Sunny (Outlook) & Temperature

$$IG = 0.5710$$

Now Sunny (Outlook) & Wind

$$IG = 0.0200$$

Here Humidity | Sunny is the highest.

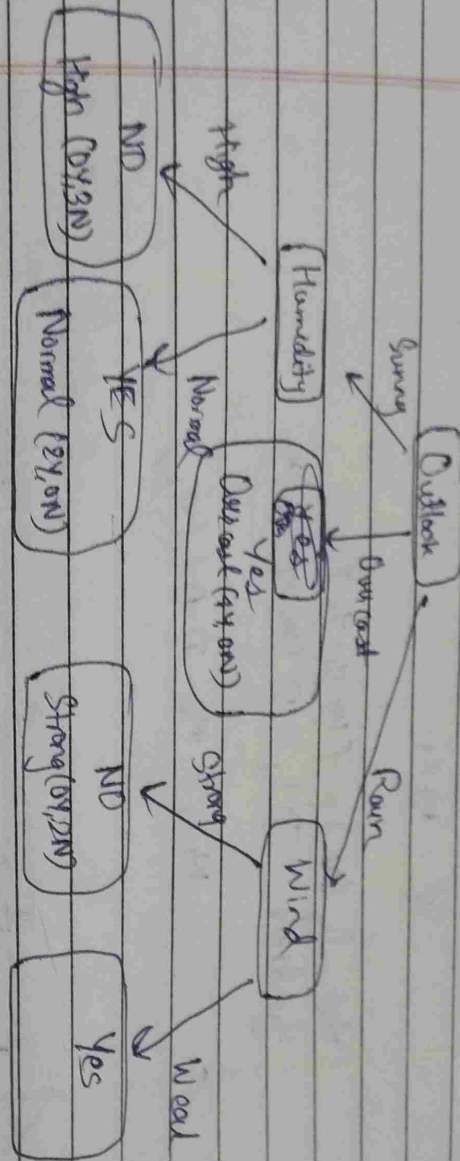


→ Now for Rain, Wind has the highest IG

(17)

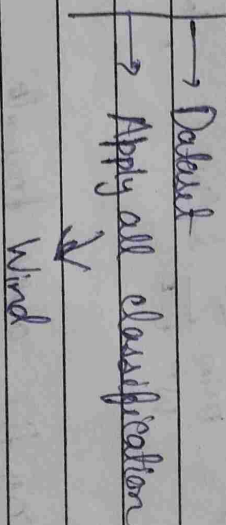


Similarly



Case Study on using

Classification Model



Online learners (Can be applied on any model)

- 1) Predict $\hat{y}$
- 2) $\hat{y} \rightarrow$ Compare true label
- 3) Learn loss
- 4) Update model

$$R_T = L_T - L_T^*$$

L_T : Cumulative loss of learning
 L_T^* : Cumulative loss fixed weight

* Gain Ratio (Temperature) = 3 values

Partitions: $|D_H| = 4, |D_M| = 8, |D_L| = 4, |D| = 14$

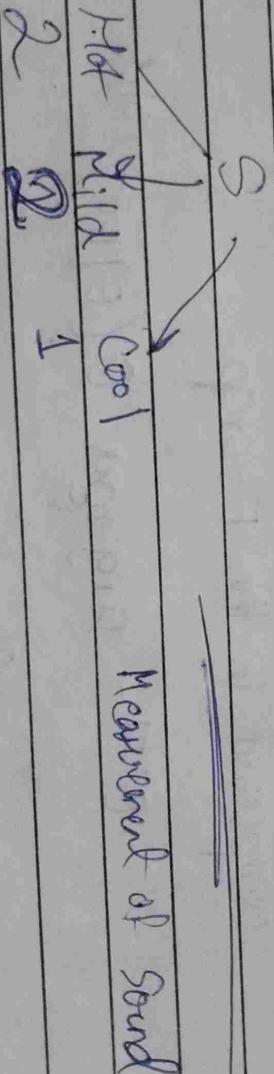
Split Info (Temperature)

$$= -\frac{4}{14} \log_2 \frac{4}{14} - \frac{6}{14} \log_2 \frac{6}{14} - \frac{4}{14} \log_2 \frac{4}{14}$$

$$= 1.5567$$

Information Gain (Temperature) = 0.0293

$$\text{Gain Ratio} = \frac{0.0293}{1.5567} = 0.0188$$



$\frac{2 \log 5}{5}$

* Expectation and Maximization algorithm

- Generative optimization method used in ML & statistics
- Find maximum likelihood of missing parameters with unknown variables.

✓ EM

Hierarchical Clustering

- Tree-like Clustering (Dendrogram)
- Bottom-Up Clustering (Agglomerative Clustering)
- Top Down Clustering (Divisive Clustering)

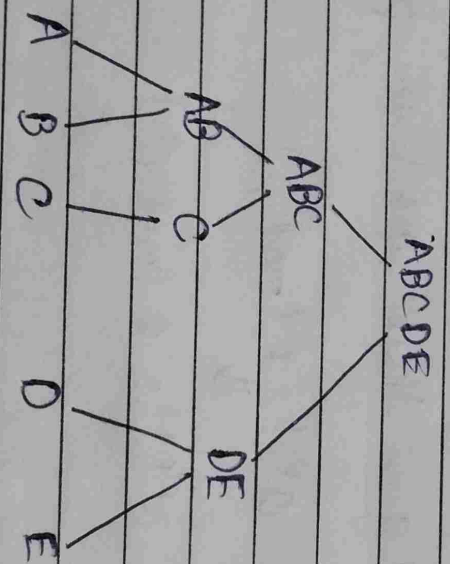
Agglomerative Clustering (Agnes)

- 1) Take every sample as a cluster
- 2) Find ~~Similarity~~ Similarity between the sample.

$$\min(\text{Distance}) = \max(\text{Similarity})$$

- 3) Merge the sample who have maximum similarity (minimum distance)

- 4) Repeat 2 and 3.



~~For step 3,~~

In step 3, we use the concept of Linkage Node

- Single Linkage (Min distance)
- Complete Linkage (Max distance)

$$\sqrt{9+4} = \sqrt{13}$$

Ham distance $\Rightarrow$ Min Similarity
 Min distance $\Rightarrow$ Max similarity

$$A = (1, 1)$$

$$B = (2, 1)$$

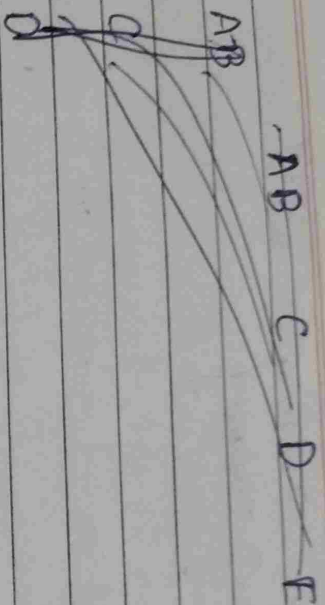
$$C = (4, 3)$$

$$D = (5, 4)$$

$$E = (2, 4)$$

	A	B	C	D	E
A	0	1	3.61	5	3.16
B	1	0	2.83	4.24	3.0
C	3.61	2.83	0	1.41	2.24
D	5	4.24	1.41	0	3.61
E	3.16	3.0	2.24	3.61	0

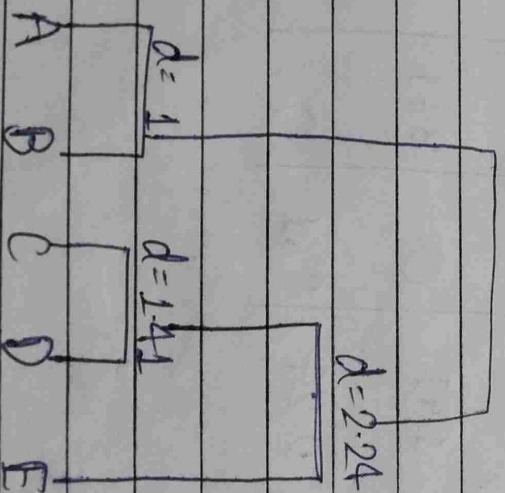
	AB	C	D	E
AB	0	$\min(3.61, 2.83) = 2.83$	4.24	3.0
C	2.83	0	1.41	2.24
D	4.24	1.41	0	3.61
E	3.0	2.24	3.61	0



	AB	CD	E
AB	0	2.83	3.0
CD	2.83	0	2.24
E	3.0	2.24	0

	AB	CDE
AB	0	2.83
CDE	2.83	0

$$d = 2.83$$



DIANA

Divisive Analysis (Top-Down)

- 1) Every sample form a ~~sample~~ large ~~cluster~~ single cluster.
- 2) Split using pairwise similarity (max-distance)

Agglomerative

Divisive

- | | |
|------------------------------|---|
| ① Every sample is 1 cluster. | ① Every 1 Cluster contain all sample |
| ② Find min dis ⁿ | ② Find max ^m distance |
| ③ Merge | ③ Split |
| ④ Bottom-Up $O(n^2)$ | ④ Top Down ($O(n^2)$) ($O(n^2)$) $O(2^n)$ |
| ⑤ Less Accuracy | ⑤ More Accurate. |